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RESEARCH ARTICLE

Immunogold Labeling to Detect *Streptococcus pyogenes* Cas9 in Cell Culture and Tissues by Electron Microscopy

María José Ulloa-Navas,[1] Patricia García-Tárraga,[1] José Manuel García-Verdugo,[1] and Vicente Herranz-Pérez[1,2,*]

Abstract

The CRISPR-Cas9 system is a powerful and yet precise DNA-editing tool in rapid development. By combining immunogold labeling and electron microscopy with the novel CRISPR-Cas9 system, we propose a new method to gain insight into the biology of this tool. In this study, we analyzed different Cas9-induced systems such as HEK293T cell line, murine oligodendrocyte progenitor cells, brain and liver to detect Cas9 expression by immunoelectron microscopy. Our results show that while Cas9 expression could be found in the nuclei and nucleopores of transfected HEK293T cells, in transfected oligodendrocyte precursor cells, Cas9 was found in cytoplasmic vesicles. In Cas9 constitutively expressing oligodendrocyte precursors, the enzyme was located in the cytoplasm of nondividing cells. Finally, while in the liver Cas9 was detected in different cell types, in the brain we found no specifically labeled cells. In conclusion, immunoelectron microscopy opens a new spectrum of opportunities to study the CRISPR-Cas9 system in a more precise manner.

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Introduction

Genome editing has become one of the most studied fields among the biomedical sciences in the last five decades. The number of applications in genome engineering have skyrocketed because of CRISPR-Cas systems.[1] First and foremost, CRISPR-Cas systems were described as an adaptive immune system in bacteria and archaebacteria against viruses and plasmids.[2,3] As an application for eukaryotic cells, CRISPR-Cas9 was originally described as a powerful tool to modify specific genome loci.[4] Nonetheless, today CRISPR-Cas systems are used not only to edit the genome and epigenome, but also to target, trace, and regulate its expression.[5]

The effectiveness of CRISPR-Cas systems to perform any activity in eukaryotic cells resides in different factors, such as the metabolic state of each cell type, which determines the biosynthesis and availability of the enzyme within the cell, and the stage of the cell cycle in which the enzyme is synthesized.[6–8] Another aspect to take into consideration is the ability of each cell type to transport Cas9 into and out of the nucleus or the target organelle. Finally, the ability of each cell type to degrade Cas9 after a time remains to be elucidated.

Another current topic usually debated in this field is the optimization of the transfection method, and how the CRISPR system *per se* can modify the characteristics of any cell type, especially in the case of stem cells or primary cultures, implying one of the most studied issues among the field.[9–11] Although high transfection efficiencies have been achieved in a wide variety of stem cells,[12–16] a gold standard to analyze not only transfection efficiency as a molecular parameter, but also the fitness status of cells in culture, has not been yet established.

Recently, some CRISPR-Cas applications to target extranuclear DNA in different organelles, such as the mitochondria, have been developed.[17] Furthermore, state-of-the-art technology nowadays allows this system to be directed to different compartments within the nucleus.[18] Therefore, it is important to develop a high-resolution tool to track Cas9 at a subcellular localization level with high precision.

In this study, we present transmission electron microscopy (TEM) and immunogold (IG) labeling, as a method to analyze the above-mentioned criteria. TEM is a classic technique that presents advantages such as a high-definition resolution at a single-cell level. When combined

[1]*Laboratory of Comparative Neurobiology, Cavanilles Institute of Biodiversity and Evolutionary Biology, University of Valencia, CIBERNED, Paterna, Spain; and* [2]*Predepartamental Unit of Medicine, Faculty of Health Sciences, University Jaume I, Castelló de la Plana, Spain.*

*Address correspondence to: *Vicente Herranz-Pérez, PhD, Laboratory of Comparative Neurobiology, Cavanilles Institute of Biodiversity and Evolutionary Biology, University of Valencia, CIBERNED, C/ Catedrático José Beltrán Martínez 2, 46980 Paterna (Valencia), Spain,* E-mail: vicente.herranz@uv.es

with IG labeling, it permits the subcellular location and distribution of a protein of interest to be observed.[19,20] Since IG-TEM can be used to study both tissues and cell cultures, it can be considered as a valid tool to detect Cas9 at a single-cell or single-organelle level and to evaluate the ultrastructural characteristics of modified cells. Therefore, here we extensively describe a method to detect *Streptococcus pyogenes* Cas9 (SpCas9) by IG labeling and TEM detection, both in Cas9-expressing cell cultures (HEK293T cell line and oligodendrocyte precursor cells [OPCs]) and in mouse tissues (liver and brain), comparing different approaches to deliver the enzyme, such as lipofection and transgenic tools.

Materials and Methods

Unless otherwise stated, all reagents were purchased from Sigma–Aldrich (St. Louis, MO).

HEK293T culture

HEK293T cells (ATCC® CRL-3216™) were seeded at a density of 4.2×10^4 cells/cm^2 in 13 mm glass coverslips on 24-well plates or 8-well Nunc Lab-Tek permanox chamber slides coated with 10 μg/mL laminin. Cells were grown in Dulbecco's modified Eagle's medium (DMEM) containing 2 mM L-glutamine, 1 × antibiotic–antimycotic solution, 5% fetal bovine serum (FBS; all from Gibco, New York, NY) at 37°C, 5% CO_2.

OPC isolation

OPCs were isolated from P0 C57BL6/J (Charles River Laboratories, Wilmington, MA) or P0 B6J.129(Cg)-Igs2tm1.1(CAG-cas9*)Mmw/J, also known as H11-Cas9 (The Jackson Laboratory, Bar Harbor, ME) mouse brains in order to isolate wild-type or Cas9 endogenously expressing OPCs, respectively. The H11-Cas9 mouse line has been previously used for pre- and post-mitotic brain-cell editing by Nishiyama *et al.*[21] The telencephalon was dissected in ice-cold Hank's Balanced Salt Solution (HBSS) with Ca^{2+} and Mg^{2+} (Gibco). Brains were transferred to 50 mL Falcon tubes (Thermo Fisher Scientific, Waltham, MA) in 10 mL ice-cold HBSS with Ca^{2+} and Mg^{2+} and subsequently incubated at 37°C for 5 min in papain solution, consisting of 0.09 mg/mL papain (Worthington Biochemical, Lakewood, NJ) and 0.02 mg/mL EDTA (Gibco) in 1× HBSS without Ca^{2+} and Mg^{2+} (Gibco). Tissue was dissociated with sterile scissors and repeated pipetting through 1,000 μL pipette tips. The papain was neutralized using a media consisting of 5% FBS and 1× antibiotic-antimycotic solution in DMEM (all from Gibco). The resulting cell suspension was filtered through a 70 μm mesh (BD Biosciences, Franklin Lakes, NJ) and centrifuged at 300 *g* for 10 min. Then, we isolated the OPC fraction using PDGFRα-MACS (Miltenyi Biotech, Bergisch Gladbach, Germany), as recommended by the manufacturer. Following this, cells were seeded at a density of 5.25×10^5 cells/cm^2 in laminin-coated 13 mm glass coverslips or on 8-well permanox chamber slides and cultured in DMEM medium containing 1×N-2 and B-27 supplements (Gibco), 10 ng/μL bFGF, and 20 ng/μL PDGF-AA (PeproTech, Rocky Hill, NJ). Medium was replaced the next day and then every other day until cells were subcultured for transfection. This process was reviewed and approved by the Animal Welfare Ethical Review Board of the University of Valencia (2017/VSC/PEA/0017).

Cell-culture transfection

For Cas9 expression, we used pSpCas9(BB)-2A-GFP plasmid (PX458), a kind gift from Dr. Feng Zhang (Addgene plasmid #48138). For green fluorescent protein (GFP) expression, we used pmaxGFP (Lonza, Basel, Switzerland). Plasmids were purified using a GenJet MiniPrep Kit (Thermo Fisher Scientific).

OPCs were transfected after 13 days *in vitro* using 750 ng of plasmid with Lipofectamine LTX with Plus reagent (Invitrogen, Carlsbad, CA), as recommended by the manufacturer, in each well of 24-well plates or with 375 ng in each well of 8-well chamber slides. HEK293T cells were transfected using 500 ng of plasmid in each well of 24-well plates or 250 ng of plasmid in each well of 8-well chamber slides, as described by Ran *et al.*,[22] with Lipofectamine 2000 (Invitrogen), as recommended by the manufacturer.

Sample processing for immunofluorescence and IG labeling

Cell-culture fixation. Cells were rinsed with 0.1 M phosphate buffer (PB) at 37°C and fixed with 4% paraformaldehyde (PFA) in 0.1 M PB at 37°C for 10 min. Then, the cells were further incubated at 4°C for 50 min. Finally, fixative solution was removed, and cells were gently washed with 0.1 M PB.

Tissue fixation and processing. P60 H11-Cas9 mice ($n = 6$) and P60 C57BL6/J ($n = 3$) mice were perfused using 0.9% saline solution at 37°C for 5 min using a flow rate of 4.5 mL/min. Then, the animals were fixed by transcardial perfusion with 4% PFA in 0.1 M PB at 37°C for 15 min. The liver and the brain were extracted and post fixed in the same solution overnight. This process was reviewed and approved by the Animal Welfare Ethical Review Board of the University of Valencia (2017/VSC/PEA/0017). For immunofluorescence (IF) detection, the organs were dehydrated and embedded in

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paraffin. Following this, 5 μm sections were obtained with an HM 340E microtome (Microm International GmbH, Walldorf, Germany). For immunoelectron detection, 50 μm sections were obtained using a Leica VT1000S vibratome (Leica Biosystems, Wetzlar, Germany).

IF detection

Cells and sections were incubated in 1:200 Immunosaver (Electron Microscopy Sciences, Hatfield, PA) in water at 60°C for 30 min. Peroxidase blocking was performed using a solution of 10% methanol and 10% H_2O_2 in 0.1 M PB. For permeabilization, the samples were washed three times for 5 min in PTA solution: 0.1% Triton X-100, 1 mg/mL bovine serum albumin (BSA) in 0.1 M phosphate buffer saline (PBS). The samples were then incubated in blocking solution (10% casein, 5% normal goat serum in PTA) for 1 h at room temperature. Subsequently, the samples were incubated overnight in primary antibodies diluted in blocking solution (1:300 rabbit-anti-Olig2, Chemicon, Temecula, CA; 1:100 mouse-anti-SpCas9, Abcam, Cambridge, United Kingdom; or 1:1000 rabbit-anti-GFAP for cell culture, DAKO, Jena, Germany). The following day, samples were thoroughly washed in PTA, and incubated in secondary antibody solution (1:500 AlexaFluor 488 goat-anti-mouse, Invitrogen; or 1:500 AlexaFluor 555 goat-anti-rabbit, Invitrogen). Samples were then washed in 0.1 M PB and incubated in 1:1,000 DAPI in water for 5 min. Finally, the samples were mounted with FluorSave™ (Merck Millipore [Calbiochem], Burlington, MA).

IG labeling for cell culture

Cell cultures cultured on 8-well permanox chamber slides were permeabilized with 0.1% Triton X-100 in 0.1 M PB for 10 min. Following this, the cells were incubated in a blocking solution consisting of 0.3% BSAc (Aurion, Wageningen, the Netherlands), 0.05% sodium azide in 0.1 M PB for 1 h. Subsequently, the cells were incubated in primary antibody (1:50 mouse-anti-SpCas9, Abcam; 1:150 rabbit-anti-Olig2, Millipore; or 1:200 chicken-anti-Olig2, Aves Labs, Tigard, OR) in blocking solution overnight at 4°C. The samples were rinsed in 0.1 M PB and then incubated in secondary antibody blocking solution consisting of 0.5% BSAc (Aurion), 0.025% CWFS gelatin (Aurion), 0.05% sodium azide in 0.1 M PB for 1 h, followed by incubation in secondary antibody (1:50 goat-anti-mouse IgG gold ultrasmall; Aurion) diluted in the same solution overnight at 4°C. To enhance gold labeling, we performed silver enhancement (R-GENT SE-LM, Aurion) for 15–25 min in the dark, followed by gentle washing in 2% sodium acetate and incubation in gold toning solution (0.05% gold chloride in water) for 10 min. The samples were then washed twice with 0.3% sodium thiosulfate in water. Finally we post fixed with 2% glutaraldehyde (Electron Microscopy Sciences) in 0.1 M PB for 30 min. Samples were rinsed and kept in 0.1 M PB containing 0.05% sodium azide at 4°C until resin embedding.

IG labeling for tissues

Processing for tissue was performed in a similar manner as for cell cultures, with a few changes. Briefly, the permeabilization step was performed by repeated freeze–thaw cycles. Before performing this process, the sections were cryoprotected in a solution containing 25% saccharose in 0.1 M PB for 30 min, after which the sections were immersed in −60°C 2-methylbutane and rapidly transferred to a room temperature saccharose solution. This cycle was repeated three or four times. Subsequently, tissues were left in 0.1 M PB and then incubated in primary antibody blocking solution for 1 h, followed by primary antibody in blocking solution (1:100 mouse-anti-SpCas9; Cell Signaling Technology, Danvers, MA) for 72 h at 4°C. The rest of the process was identical to that described for cell cultures.

Sample processing for TEM

Samples were embedded in resin, as previously described.[20] Briefly, samples were post fixed with 1% osmium tetroxide (Electron Microscopy Sciences), 7% glucose in 0.1 M PB for 30 min at room temperature, washed in deionized water, and partially dehydrated in 70% ethanol. Afterwards, the samples were contrasted in 2% uranyl acetate (Electron Microscopy Sciences) in 70% ethanol for 2 h at 4°C. The samples were further dehydrated and infiltrated in Durcupan ACM epoxy resin at room temperature overnight, and then at 60°C for 72 h. Once the resin was cured, immunolabeled sections were selected and cut into ultrathin sections (60–80 nm) using an Ultracut UC6 ultramicrotome (Leica Biosystems). These sections were placed on Formvar-coated single-slot copper grids (Electron Microscopy Sciences) stained with lead citrate and examined at 80 kV on a FEI Tecnai G^2 Spirit (FEI Company, Hillsboro, OR) transmission electron microscope equipped with a Morada CCD digital camera (Olympus, Tokyo, Japan).

Statistical analysis

Quantitation of transfected OPCs and HEK293T cells was performed in triplicate experiments by counting the number of IF-positive cells relative to the total number of cells in three different fields (Supplementary Fig. S1A–E). Statistical analyses were performed using GraphPad Prism v8 (GraphPad, San Diego, CA).

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Differences among different time points were assessed by one-way analysis of variance followed by Tukey's *post hoc* test. All numerical data are expressed as the mean ± standard error of the mean. All statistical tests were two-sided, and a p-value of <0.05 was considered to be statistically significant.

Results

We evaluated immunoelectron microscopy as a useful technique to detect Cas9 precisely in different biological systems, including cell cultures and tissue sections (Fig. 1). First, to standardize our method in cell lines, we used transfected human embryonic kidney 293 cell line (HEK293T). In addition, we explored the expression and immunodetection of Cas9 in OPC primary cultures from the mouse brain. Finally, we used a similar strategy to analyze the liver and brain of transgenic mice that constitutively express SpCas9.

Immunoelectron microscopy improves the visualization of the subcellular distribution of SpCas9 in HEK293T cell line

To identify the strengths and flaws of our method, we first transfected HEK293T cells in order to assess the differences between Cas9 IF and IG detection (Fig. 1). In every case, IF was used as a control to evaluate Cas9 expression at the subcellular level. First, we transfected the cells with PX458 plasmid containing SpCas9 and GFP sequences. Subsequent IF detection 24 h after transfection showed that both SpCas9 and GFP were present within the nucleus and in the cytoplasm of the cells (Fig. 2A and Supplementary Fig. S1A; 43.94 ± 2.75% GFP^+ cells; 40.58 ± 2.99% $SpCas9^+$ cells; $n = 1{,}650$ cells). However, IG labeling displayed a finer localization of the enzyme (Fig. 2B–F). By using this method, we observed that cells expressed SpCas9 protein in the cytoplasm, although we could not detect it inside any particular organelle (Fig. 2C). A small fraction of cells expressed SpCas9 exclusively in the nucleus but not in the nucleolus (Fig. 2D), and interestingly enough, IG labeling allowed us to detect Cas9 clearly in the boundary of many nuclear pores, suggesting a potential nuclear translocation of this protein (Fig. 2E and F).

Together, these results indicate that immunoelectron microscopy is a valuable tool for gaining insight into the subcellular dynamics of Cas9.

Immunoelectron microscopy is a valid method for tracking SpCas9 in transfected primary cells *in vitro* over time

To track SpCas9 at different times after transfection, we introduced PX458 plasmid in OPCs and fixed the cultures after 10 h, 24 h, 72 h, and 5 days (Fig. 3A–L). Then, we performed IF and IG-TEM detection to identify the subcellular localization and dynamics of SpCas9 over time. At 10 h, we did not detect SpCas9 by either IF or IG (Fig. 3A–C). This was also the case for GFP after IF detection (Fig. 3A), suggesting that the plasmid was not expressed at the shortest time point analyzed in our study. Interestingly, 24 h after transfection, the label

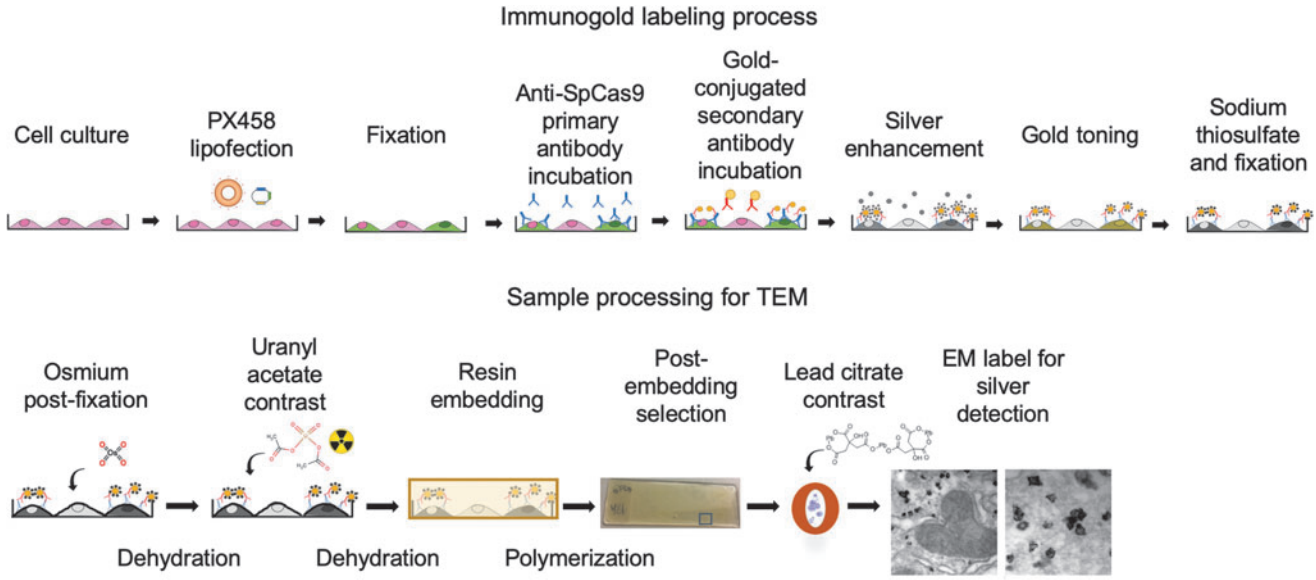

FIG. 1. Step-by-step diagram describing the procedure followed from cell culture to transmission electron microscopy analysis. See Materials and Methods.

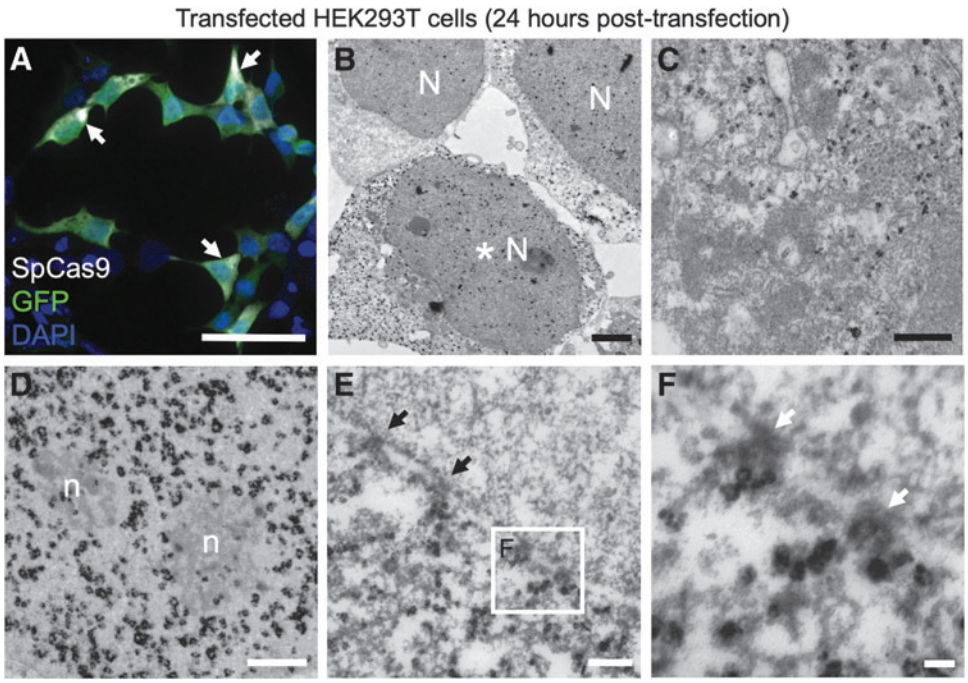

FIG. 2. Expression of green fluorescent protein (GFP) and *Streptococcus pyogenes* Cas 9 (SpCas9) in HEK293T cells transfected with PX458 24 h after transfection. **(A)** Immunofluorescence (IF) analysis of transfected HEK293T cells shows SpCas9 expression broadly distributed in the nucleus and cytoplasm (white arrows). Interestingly, GFP tag does not perfectly co-localize with SpCas9. **(B)** SpCas9 immunoelectron detection (immunogold [IG]) results in cells with different amounts of label. Intensely labeled cells, with signal in both the nucleus and cytoplasm (asterisk), contrasts with cells presenting a less abundant labeling. **(C)** Detail of the cytoplasm of a SpCas9-labeled cell. **(D)** High-power image of the nucleus of a SpCas9-labeled cell. Remarkably, the label is found throughout the nucleus but not in the nucleolus. **(E)** Gold-particle labeling can be frequently found associated to nuclear pores (black arrows). **(F)** High-magnification photomicrograph showing the presence of labeling in the nuclear pore boundary (white arrows). N, nucleus; n, nucleolus. Scale bars: **(A)** 100 μm; **(B)** 1 μm; **(C)** 500 nm; **(D)** 1 μm; **(E)** 200 nm; and **(F)** 50 nm.

was detected by IF and IG in small cytoplasmic vesicles surrounding the nucleus (Fig. 3D–F) and in the perinuclear space only detectable by IG (Fig. 3E and F). Intriguingly, at this time point, we were able to detect only one cell in which Cas9 presented a nuclear localization (Fig. 4A), indicating that the lack of nuclear label in the majority of the cells is not due to a methodological limitation. IF analysis of the cells 72 h post transfection suggested that Cas9 was expressed in specific areas of the cytoplasm (Fig. 3G). We corroborated this observation by IG-TEM, which allowed us to determine that Cas9 compartmentalized into small cytoplasmic vesicles (Fig. 3H and I). To investigate the role of the vesicles that packed the enzyme in this cell type further, we tracked

FIG. 3. Expression of GFP and SpCas9 in oligodendrocyte precursor cells (OPCs) transfected with PX458. **(A–C)** No expression of SpCas9 is detected 10 h after transfection by **(A)** immunofluorescence or **(B** and **C)** immunogold transmission electron microscopy (IG-TEM). **(D, G,** and **J)** SpCas9 immunofluorescent labeling in the cytoplasm of OPCs after 10 h, 24 h, 72 h, and 5 days upon transfection (white arrows). **(E, H,** and **K)** IG followed by TEM images showing SpCas9 labeling within small vacuoles or vesicles 24 and 72 h after SpCas9 transfection (black arrows in **E** and **H**), while 5 days post transfection, the label is confined to dense vacuoles with lysosomal morphology (black arrows in **K**). **(F, I,** and **L)** High-magnification photomicrographs of the vacuoles and lysosomes (white arrows) where the label is detected 24 h, 72 h, and 5 days post transfection, respectively. **(M** and **N)** Quantification of **(M)** GFP- or **(N)** SpCas9-positive cells 10 h, 24 h, 72 h, and 5 days after transfection. Data shown are mean ± standard error of the mean. ** indicates a statistical p-value < 0.01. *** indicates a statistical p-value < 0.001. Scale bars: **(A, D, G,** and **J)** 10 μm; **(B, E, H,** and **K)** 1 μm; **(C, F,** and **I)** 500 nm; **(L)** 200 nm.

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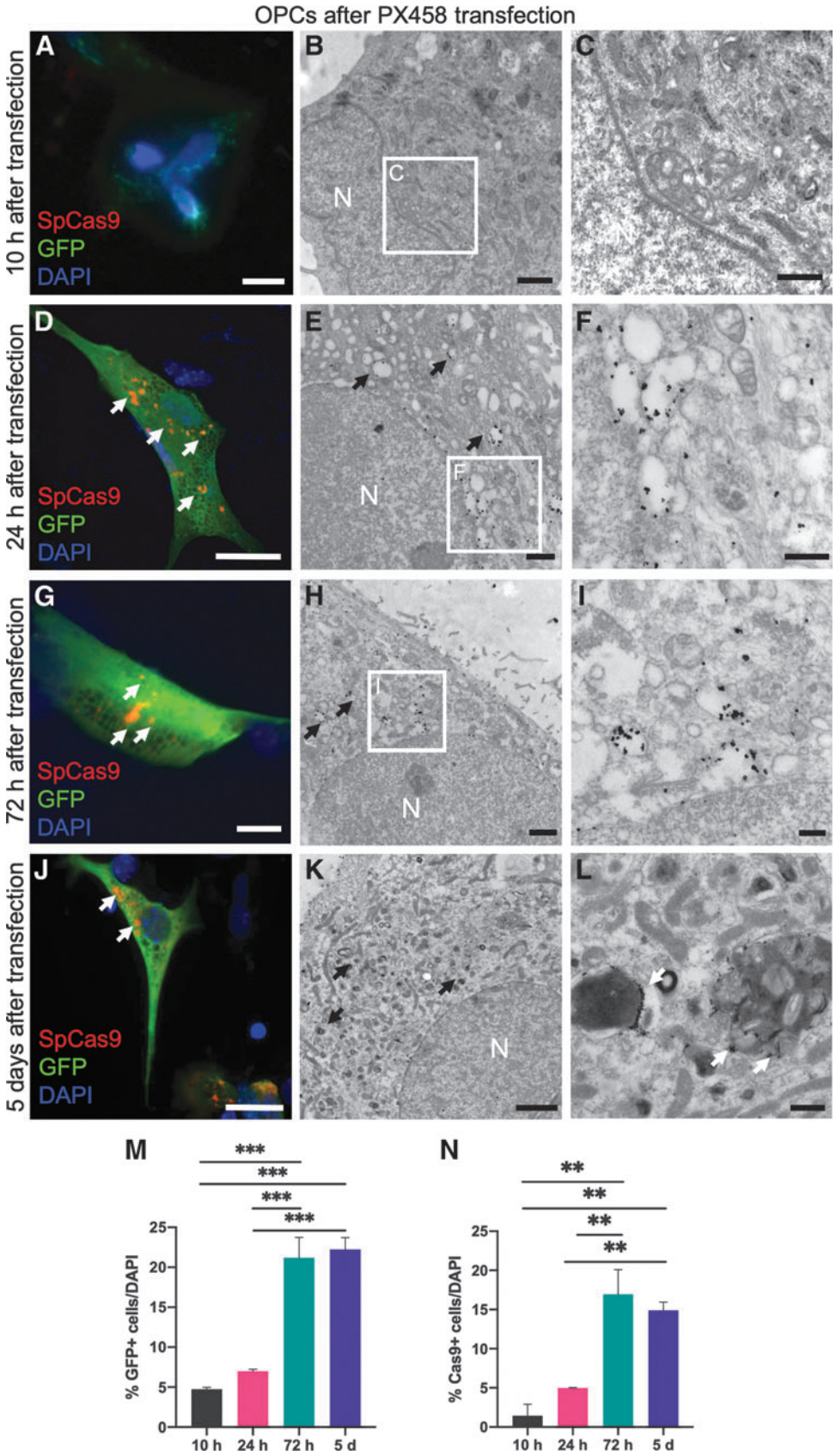

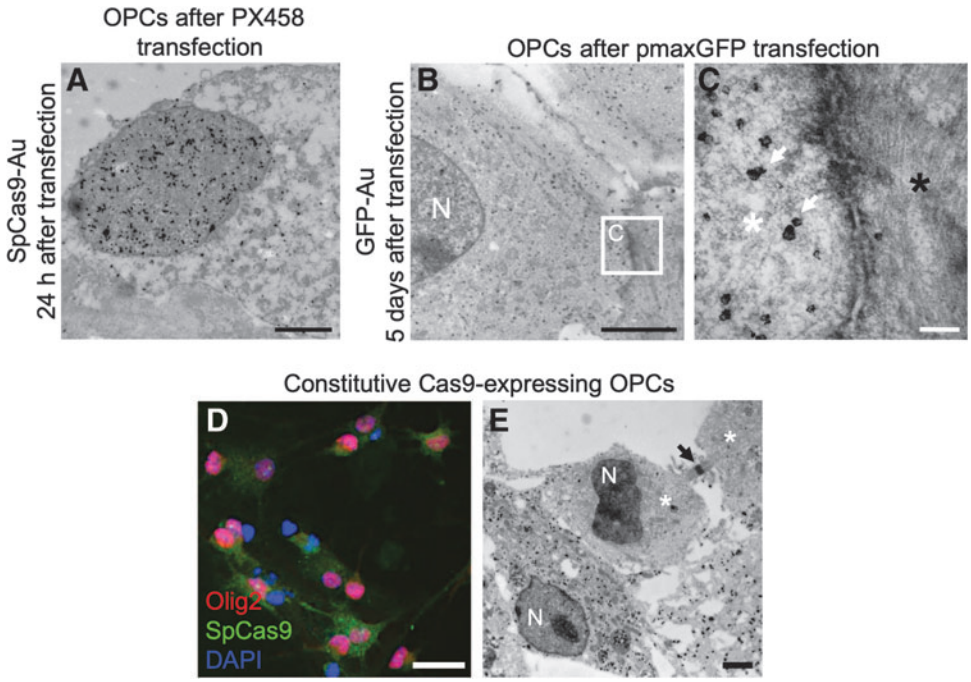

FIG. 4. SpCas9 is distinctly localized in OPCs. **(A)** A minority of OPCs display nuclear localization of SpCas9. **(B)** GFP IG-labeling in pmaxGFP-transfected OPCs. The expression of this plasmid does not induce the formation of vacuoles or GFP vesicle packaging. **(C)** High-magnification photomicrograph showing the cytoplasm of a GFP-labeled OPC (white asterisk) and a non-labeled OPC (black asterisk) showing no vacuoles. Gold particles are highlighted by white arrows. **(D)** IF analysis of OPCs derived from H11-Cas9 mice showing SpCas9 expression the cytoplasm. **(E)** TEM analysis reveals SpCas9 label widely distributed in the cytoplasm of OPCs. This labeling is not confined to vesicles. Interestingly, cells going through cell division do not express Cas9 (asterisk). In this photomicrograph, a mitotic cell in telophase can be observed. The black arrow points to the mitotic spindle. Scale bars: **(A)** 1 μm; **(B)** 5 μm; **(C)** 500 nm; **(D)** 25 μm; **(E)** 1 μm.

SpCas9 5 days after transfection. At this time point, we observed by IF that SpCas9 was still present in discrete regions of the cytoplasm, as we observed at 24 and 72 h (Fig. 3D, G, and J). By combining IG labeling and TEM, we detected that the enzyme localized to electron-dense lysosome-like structures and also autophagosomes, which displayed mitochondria and membranous structures in their content (Fig. 3K and L).

Quantitative analysis of GFP^+ and $Cas9^+$ cells relative to the total number of cells showed an increase with time until 72 h post transfection, which was sustained at similar levels 5 days after transfection (Fig. 3M and N). We found statistically significant differences in the number of GFP^+ cells between 10 h ($4.75 \pm 0.23\%$; $n = 83$ cells) and 72 h post transfection ($21.20 \pm 2.55\%$; $n = 110$ cells, $p = 0.0002$); 10 h and 5 days post transfection ($22.25 \pm 1.47\%$; $n = 67$ cells, $p = 0.0001$); 24 h ($6.99 \pm 0.25\%$; $n = 517$ cells) and 72 h post transfection ($p = 0.0006$); 24 h and 5 days post transfection ($p = 0.0004$; Fig. 3M). The number of $Cas9^+$ cells also increased in a similar trend. We found significant differences between 10 h (1.45 ± 1.45; $n = 83$ cells) and 72 h post transfection ($16.95 \pm 3.14\%$; $n = 110$ cells, $p = 0.0013$); 10 h and 5 days post transfection ($14.90 \pm 1.05\%$; $n = 67$ cells, $p = 0.0033$), 24 h ($4.98 \pm 0.06\%$; $n = 517$ cells) and 72 h post transfection ($p = 0.0068$); 24 h and 5 days post transfection ($p = 0.0194$; Fig. 3N).

To elucidate further whether the specific packaging of Cas9 was due to a transfection issue, we transfected pmaxGFP (Lonza) into OPCs. This plasmid is used as a control for transfection when using GFP as a fluorescent marker. As this plasmid does not express Cas9, we performed IG to detect GFP. Our results indicate that GFP is extensively expressed in the cytoplasm and is not packaged into vesicles (Fig. 4B and C), supporting the notion that vesicle packaging is a specific feature of Cas9-induced expression in this cell type.

Overall, using IG-TEM provides a detailed view at a single-cell and subcellular level, allowing Cas9 to be detected, even in discrete organelles. Our results also suggest that Cas9 expression via lipofection in OPCs can result in enzyme degradation in lysosomes.

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IG-TEM favors the validation of an efficient expression method in primary cultures

One of the main issues reported by CRISPR users is the validation of the expression method in primary and hard-to-transfect cell cultures. As a proof of principle to endorse that ultrastructural analysis contributes to validate an optimal expression method for this type of cultures, we isolated cells that endogenously express SpCas9 to compare their ultrastructure to that of transfected cultures. For this purpose, we isolated OPCs from P0 H11-Cas9 mice and performed the same detection protocols as for transfected OPCs. In this case, using IF, we found that Cas9 is widely expressed throughout the cell cytoplasm (Fig. 4D). Still, IG-TEM allowed us to confirm that cells express Cas9 in the cytoplasm. Strikingly, mitotic cells, even those in cytokinesis, did not express Cas9, suggesting that OPCs undergoing cell division do not express the enzyme (Fig. 4E). Thereby, we corroborated that ultrastructural studies are significant to ascertain the basic biology of different cells expressing the CRISPR-Cas9 system.

IG enhances specificity of label detection in the liver

To set up the methodology for Cas9 IG detection and TEM processing, we performed IF and IG-TEM on liver and brain tissue of P60 H11-Cas9 mice ($n = 6$).

First, we used IF to study the expression of SpCas9 by astrocytes and oligodendrocyte progenitors within the motor cortex of the brain (Fig. 5). In the case of Olig2-positive OPCs, Cas9 was expressed within the cytoplasm. This result is consistent with our results in OPC cultures *in vitro*, indicating that Cas9 is in most cases in the cytoplasm in this specific cell type (Fig. 5E–H). To identify astrocytes, we used immunolabeling for the glial fibrillary acidic protein (GFAP) together with Cas9 immunolabeling. Using this approach, we determined that astrocytes do not express Cas9 in this animal model (Fig. 5I–L). Interestingly, we found Cas9-positive cells with neuronal morphology in the cortex of the brain, with both cytoplasmic and nuclear signal. On the contrary, using immunoelectron microscopy, we were not able to detect Cas9 expression in any brain structure after trying several different conditions.

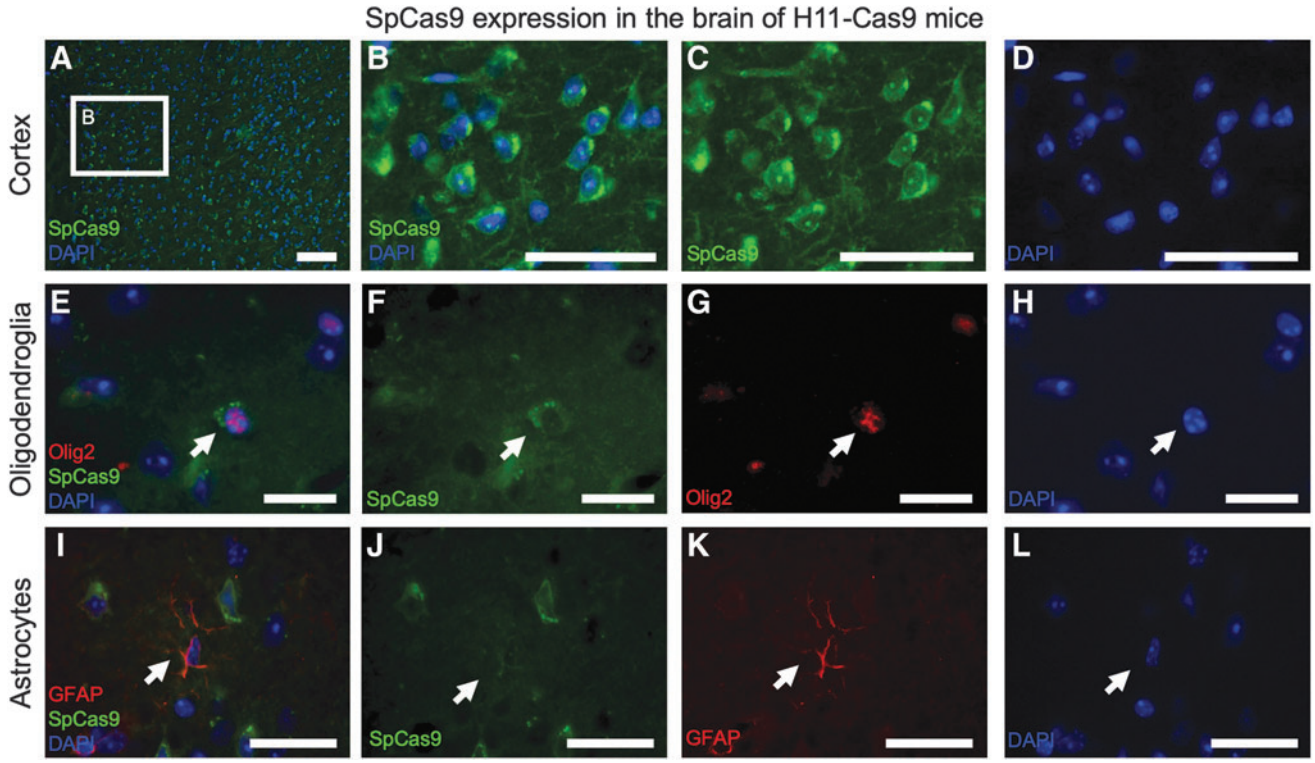

FIG. 5. Immunofluorescence detection of SpCas9 in the brain of H11-Cas9 mice. **(A–D)** Abundant SpCas9-positive cells with neuronal morphology can be observed in the brain cortex. The labeling can be observed in the cytoplasm and in specific areas of the nucleus. **(E–H)** Olig2-positive cells display a cytoplasmic SpCas9 distribution (white arrows in **E–H**), while other glial cells such as GFAP-positive astrocytes **(I–L)** do not express the enzyme (white arrows in **I–L**), indicating some variability in SpCas9 expression among cell types in this transgenic mouse line. Scale bars: **(A)** 100 μm; **(B–D)** 50 μm; **(E–L)** 10 μm.

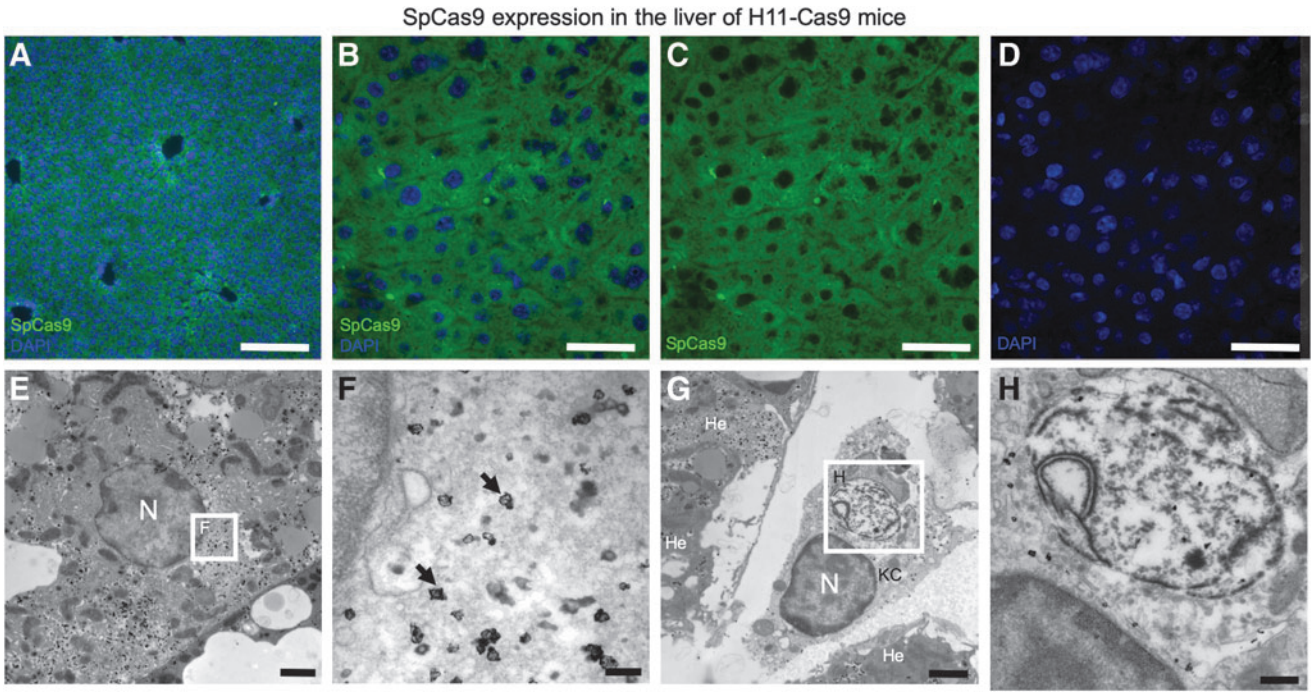

FIG. 6. Immunofluorescence detection of SpCas9 in the brain of H11-Cas9 mice. **(A–D)** IF detection of SpCas9 in the liver exhibits a ubiquitous cytoplasmic labeling in most cells. **(E)** IG-TEM analysis displays a hepatocyte presenting a homogenous distribution of SpCas9 throughout the cytoplasm. **(F)** High magnification photomicrograph of the labeled cytoplasm shown in **(E)** (arrows point to gold particles). **(G)** Kupffer cell labeled with SpCas9. **(H)** High-magnification micrograph of the labeled cytoplasm shown in **(G)**. He, hepatocyte; KC, Kupffer cell. Scale bars: **(A)** 250 μm; **(B–D)** 50 μm; **(E)** 2 μm; **(F)** 200 μm; **(G)** 2 μm; **(H)** 500 nm.

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IF of the liver displayed a wide distribution of the label across the tissue (Fig. 6A–D). Intriguingly, IG in the liver allowed us to detect the gold labeling broadly distributed in the cytoplasm of some hepatocytes and Kupffer cells (Fig. 6E–H). This is an interesting fact, given that the liver is a highly autofluorescent organ, and IG can be a tool to discard autofluorescence while providing a high-resolution image.

The fact that IG can efficiently detect Cas9 in liver but not brain tissue suggests that not all tissues may be optimal for IG processing and analysis.

Discussion

Cutting-edge studies have transformed the bacterial defense system CRISPR-Cas into a potent tool for genome editing,[3,23,24] not only for nuclear but also for extranuclear DNA.[2,4,17,25,26] However, to our knowledge, there are no studies that present ultrastructural data for CRISPR applications. Conventional techniques, such as IF Cas9 detection, do not allow for subcellular tracking, nor do they provide super-resolution images that facilitate the understanding of the dynamics of this bacterial enzyme within the eukaryotic cell.

Therefore, the purpose of this study was to develop a tool to achieve Cas9 identification and high-resolution imaging for individual cells. Accordingly, we used different *in vitro* and *in vivo* studies to assay IG combined with TEM in order to analyze different cellular features of Cas9 expressing cells.

We transfected HEK293T with PX458 Cas9 expressing plasmid as a control and proof of principle for cell-line experimentation. In this aspect, our results using IF show that 24 h after transfection, these cells display a broad distribution of the enzyme throughout the nucleus and cytoplasm. Still, IG provides a better resolution for subcellular localization. TEM analysis allows the detection of the enzyme within the nucleus but not in the nucleolus, and furthermore, it permits the visualization of Cas9 in the proximity of nuclear pores, suggesting its nuclear translocation.

As a proof of concept for Cas9 detection in primary cell cultures, we used oligodendrocyte progenitors isolated from P0 mice from control and transgenic animals that constitutively express Cas9. TEM analysis allowed us to detect Cas9 mostly in vesicles and in the perinuclear space, while this effect was not observed when these

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cells were transfected with GFP. However, recent articles achieve genome editing in OPCs by using either human-induced pluripotent stem cells together with lipofection[27] or viral vectors in primary OPCs.[28] This indicates that the method of isolating or inducing OPC production and the transfection method is rather important in this cell lineage. Our results suggest that nuclear internalization and/or enzyme availability in the nucleus of OPCs is scarce and therefore undetectable, opening the field to more studies that take into account aspects to understand and correlate the efficiency of genomic edition with nuclear availability and nuclear transportation in primary oligodendroglial lineage cell cultures.

Taking into account the broad cytoplasmic distribution of GFP in PX458 and pMAX GFP transfections, together with our quantitative analysis showing an increase in GFP expression over time, we suggest that GFP is a stable protein in OPCs. We found a similar increase in the number of Cas9-positive cells with time. Nevertheless, the subcellular distribution of this protein appeared to be restricted to the perinuclear space and cytoplasmic vesicles at the short and medium term after transfection (24 and 72 h, respectively). Furthermore, in the longer term (5 days after transfection), Cas9 appeared mostly confined to lysosomal vesicles and autophagosomes, suggesting that the enzyme is degraded by OPCs after lipofection. In contrast, Cas9-expressing OPCs displayed a wide cytoplasmic distribution of the enzyme in nondividing cells. These findings reveal the potential of immunoelectron microscopy to provide complementary information to that offered by other techniques, such as IF or Western blot, to elucidate the final fate of proteins at the subcellular level. Additionally, IG-TEM favors the comparison of transfected versus non-transfected and dividing versus nondividing cells within the same culture.

Finally, we also demonstrate that IG-TEM can be successfully used for Cas9 detection in mouse tissues such as the liver. Conversely, while IF was able to reveal Cas9 expression in the brain, IG-TEM was not effective for this purpose. While we cannot rule out the possibility that enhanced methods could improve Cas9 detection in difficult tissues, there is evidence supporting the fact that antibody–antigen interactions can be complex and unpredictable, especially in applications such as immunoelectron microscopy.[29] Technical improvements in antibody design, tissue processing, and Cas9 expression will help to overcome these limitations.

Conclusion

In conclusion, we have adapted a classic method such as TEM with a state-of-the-art tool such as the CRISPR-Cas system to potentiate this technique in the cell biology field and give a clearer vision and a more exact validation for CRISPR users. Ultrastructural studies together with IG labeling contribute to a deeper comprehension of Cas9 biology at the subcellular level. Nevertheless, further research is needed in this field to understand CRISPR-Cas9 dynamics better in a wider spectrum of cell types and tissues.

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Author Disclosure Statement

The authors declare no competing financial interests.

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Supplementary Material

Supplementary Figure S1

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